



DEALER BEST PRACTICE SERIES

Tool For Easy Measurement of Backlash Between Bevel Gear and Pinion

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

For the accurate measurement of the backlash between the bevel gear and pinion in Caterpillar wheel loaders, it is important to keep the pinion gear in a fixed position while taking measurements. This Best Practice is a simple fabricated tool to lock the pinion gear while measuring backlash. Refer to machine Service Manuals for differential bevel gear assembly, procedure to test / adjust bevel gear backlash.



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Tool For Easy Measurement of Backlash Between Bevel
Gear and Pinion

DATE
2/19/2015

CHG
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NUMBER
1205-4.03-1273

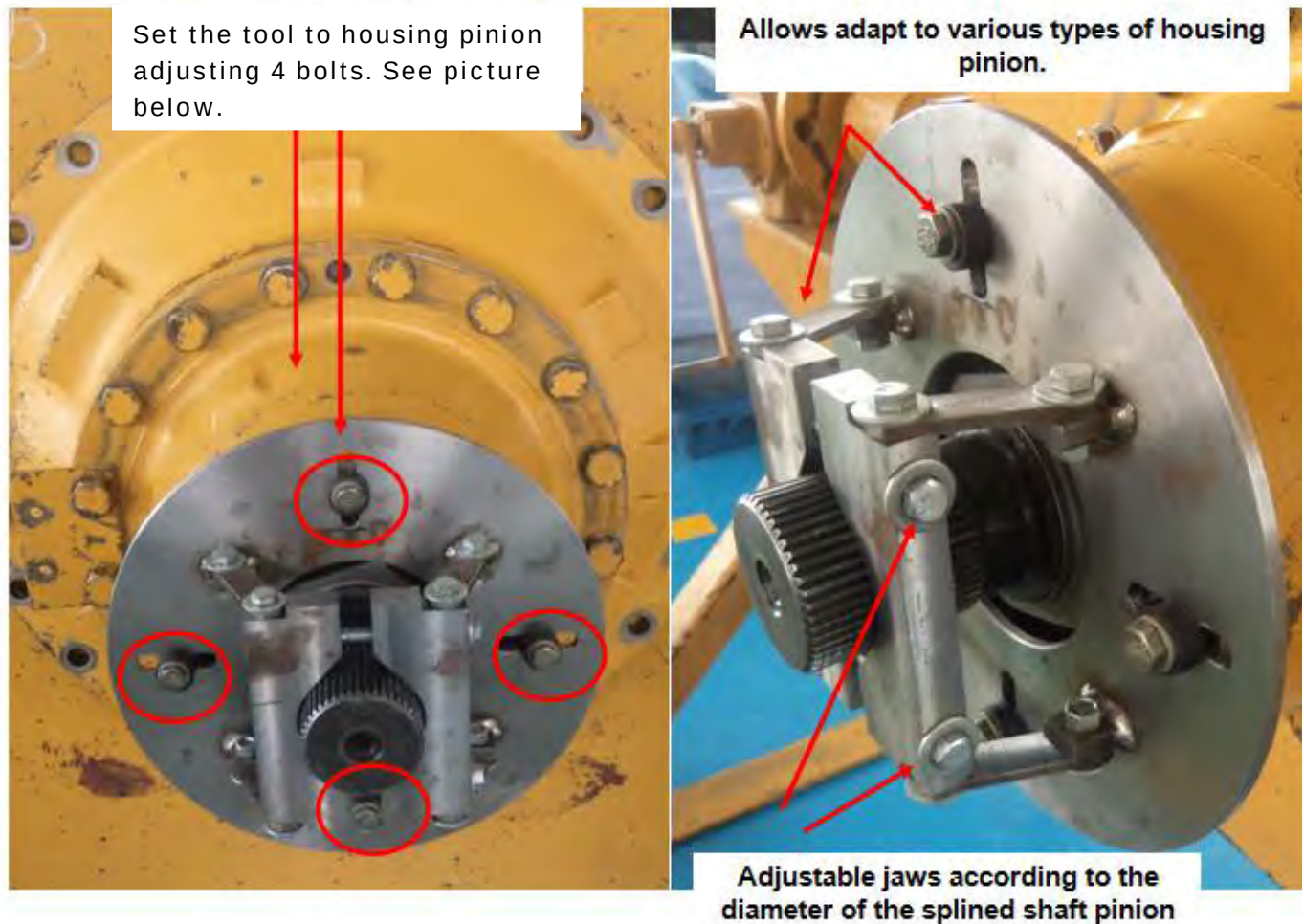
2.0 Best Practice Description

Ferreyros has developed a very simple tool to fix the pinion, for measuring the backlash of the bevel gear and pinion.

This tool allows easier measurement of the backlash without repeating the procedure several times. This tool enables accurate and reliable measurement to be performed in the least amount of time.

Process:

1. Set the tool to the housing pinion with 4 bolts.
2. Adjust the jaws according the diameter of the splined shaft pinion.
3. Tighten 2 bolts to secure jaws to the pinion.



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4. Proceed to the measurement of backlash in the bevel gear.



3.0 Implementation Steps

Design and fabricate a tool as shown above.

4.0 Benefits

- More accuracy in the measurement in less time.
- Ensure that the test procedure meets the specification CAT.
- The tool is adaptable for models 824, 824B, 824C, 834B, 834C, 844, 844C, 980G, 988B, 988F, 988G, 988H and many more.

5.0 Resources Required

- Fabrication of tool.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

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8.0 Acknowledgements

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